INFO6105 Final Project Report: Predicting Ride Demand and Revenue Loss in NYC

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URL Video :
https://www.youtube.com/watch?v=4gnVIXm6WPQ



Project Title: Predicting Ride Demand and Revenue Loss in NYC

Project Overview

This project explores how ride-sharing platforms can use historical ride data to make better operational decisions through demand forecasting and revenue loss analysis. Using January 2025 NYC Yellow Taxi trip data, the project focuses on two main questions:

1. Can we accurately forecast hourly ride demand using historical data?
2. How much revenue is potentially lost when not enough drivers are available during certain hours?

By leveraging time series forecasting and visual storytelling in R, this project reveals demand patterns. It identifies high-impact operational windows, such as peak evening hours and underserved late-night periods.

Dataset and Tools

- **Source**: NYC Taxi and Limousine Commission (TLC) Trip Record Data
- **Dataset Used**: yellow_tripdata_2025-01.csv
- **Records**: ~7 million trips

Tools & Libraries in R:

- `data.table` for efficient data loading
- `tidyverse` and `lubridate` for data wrangling and time formatting
- `forecast` for time series modelling
- `ggplot2` for data visualisation

Methodology

- Cleaned and filtered NYC taxi data to remove invalid entries (e.g., trips with zero distance or fare)
- Converted pickup timestamps to hourly format and aggregated total rides per hour
- Used ARIMA to forecast the next 24 hours of ride demand
- Created a revenue loss simulation by assuming a capped driver availability of 50 rides/hour between 12 AM – 6 AM
- Built visualisations including forecast plots, boxplots for volatility, bar charts for revenue loss, and heatmaps for hourly demand

Research Questions and Findings

Q1: Can we accurately forecast hourly ride demand?

Answer: Yes.

- The ARIMA forecast closely followed the historical trend, especially the daily peaks from 4 PM to 8 PM
- Demand was highly predictable due to clear recurring daily patterns

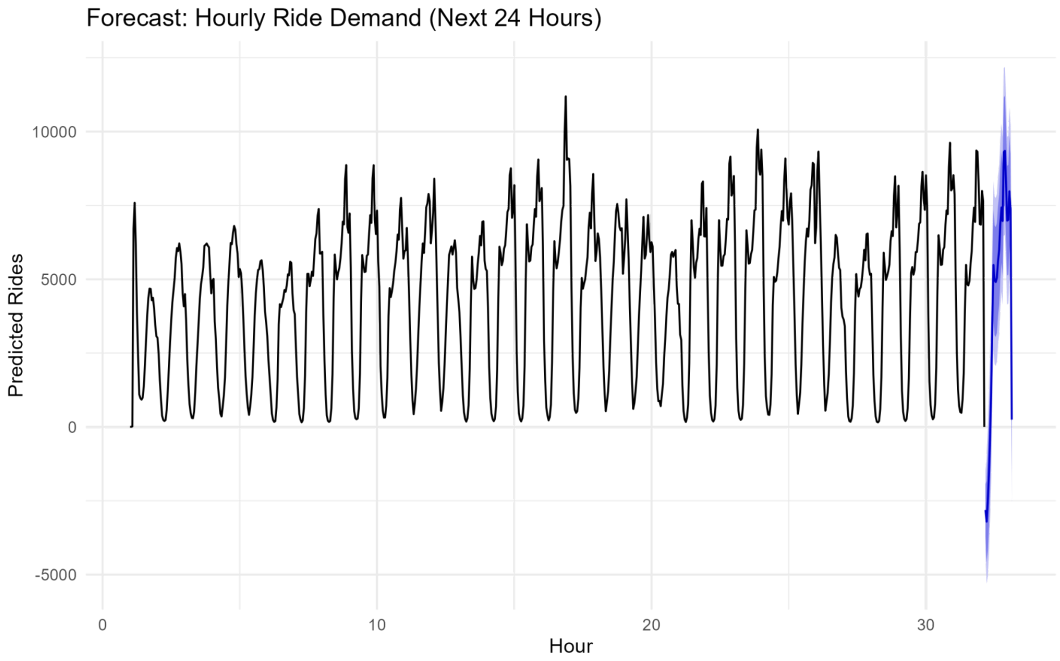
Q2: How much revenue is lost during low-driver-availability hours?

Answer: Significant revenue is lost.

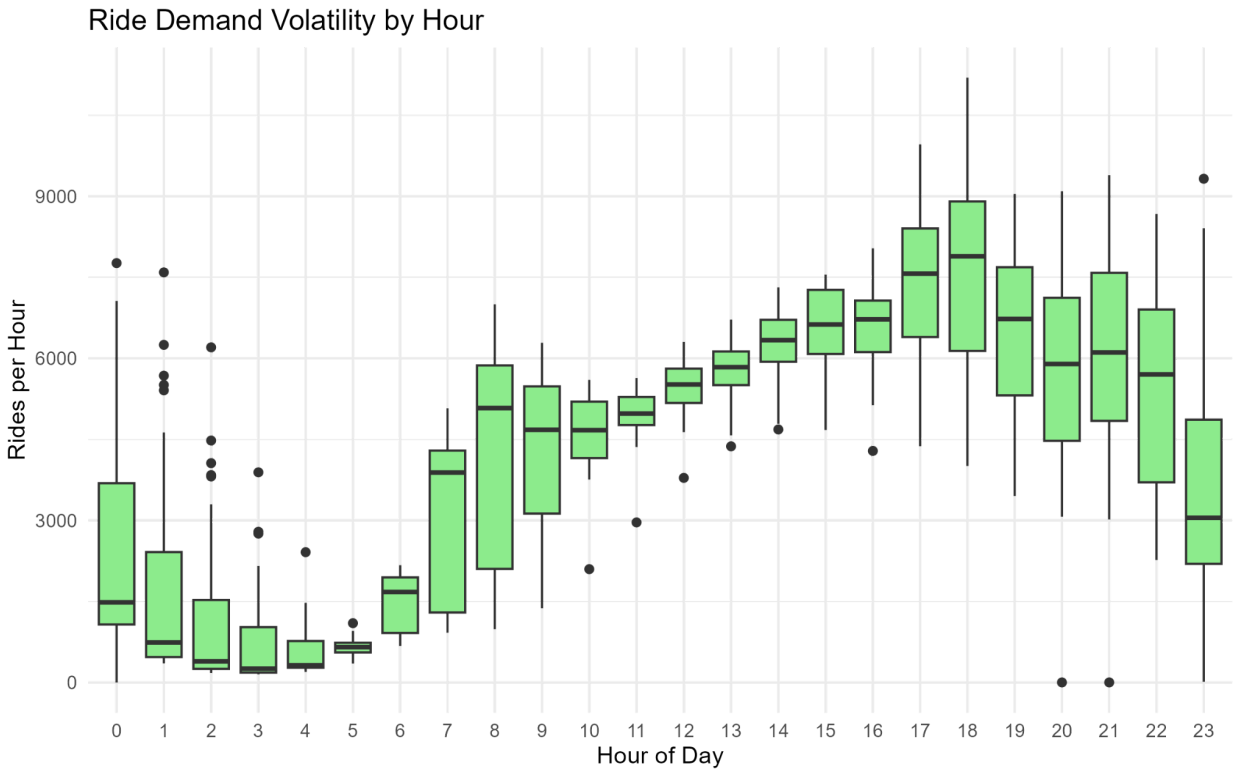
- Simulation with a 50-ride/hour cap during 12 AM–6 AM revealed estimated losses of $300–$400/hour
- Cumulative losses were highest on weekends, especially Friday and Saturday late nights

Visual Insights

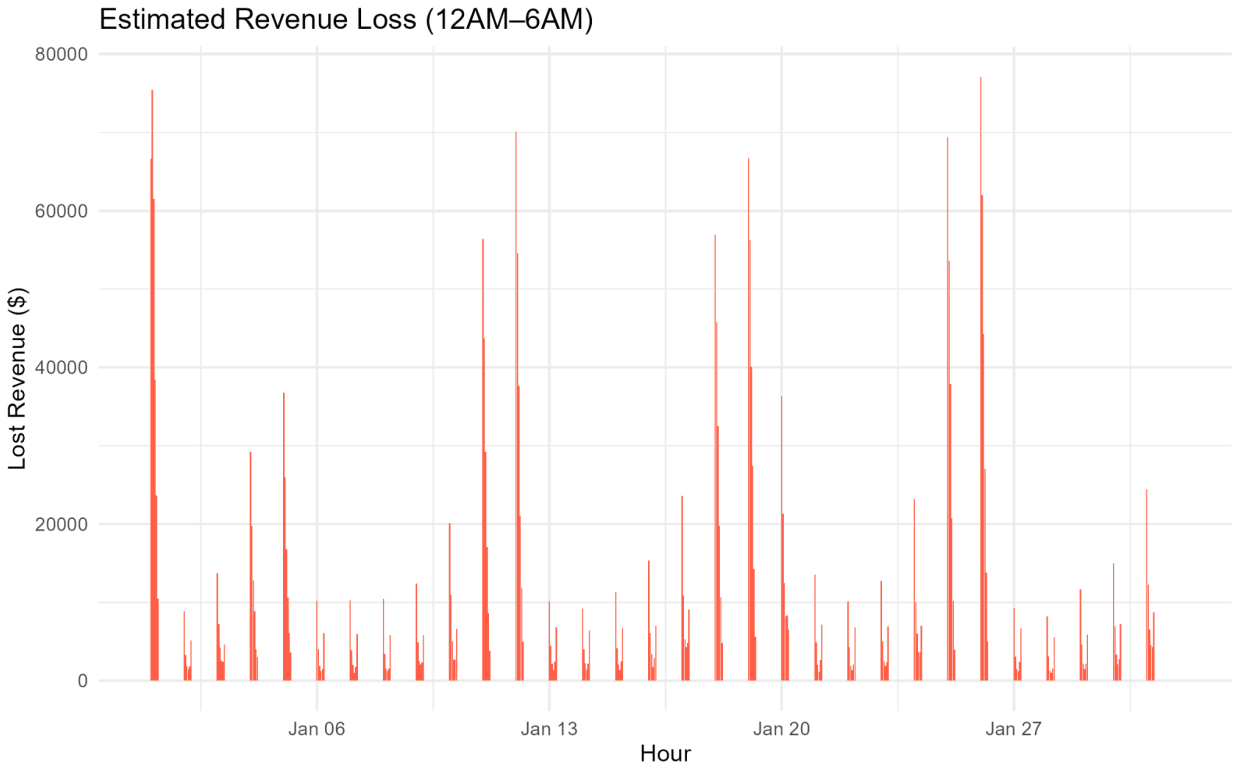
- **Forecast Plot**: Demonstrates accurate prediction of ride volume over 24 hours



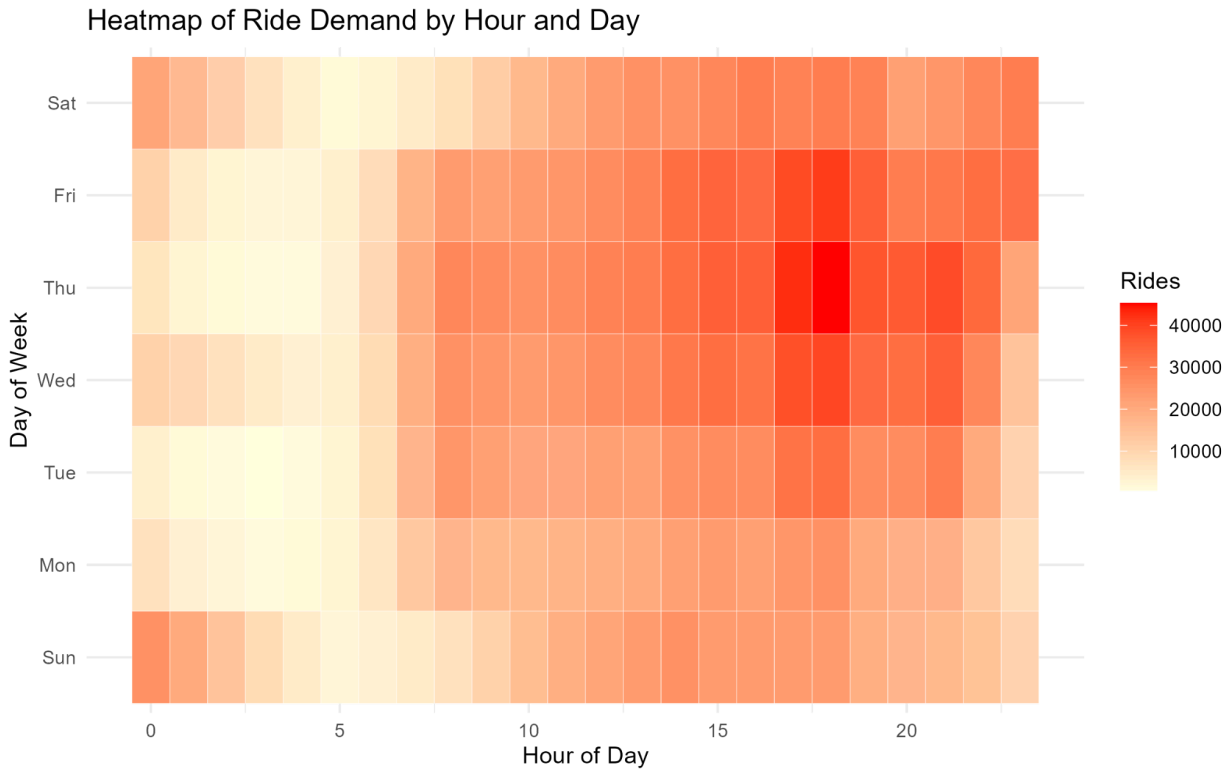
- **Boxplot**: Highlights hour-by-hour variability; evening hours are the most volatile



- **Revenue Loss Chart**: Reveals substantial financial losses due to driver unavailability



- **Heatmap**: Shows demand intensity by day and hour, with Friday and Saturday nights peaking late



Business Recommendations

- Use hourly forecasting to schedule drivers more efficiently, especially before evening peaks
- Implement incentive-based driver programs for overnight shifts (particularly on weekends)
- Use volatility data to activate surge pricing or flexible deployment plans
- Leverage heatmap insights to balance weekday vs weekend operations

Challenges Faced

- Initial ARIMA tuning and interpreting model output (e.g., confidence intervals)
- Handling missing or misaligned timestamps
- Balancing visual clarity vs chart complexity (especially in heatmaps)

Conclusion

This project demonstrates that historical ride data can be effectively used to forecast demand and assess revenue risk. By combining time series modelling with visual storytelling, ride-sharing companies can turn data into actionable decisions. The findings suggest that proactive planning during high-demand hours and targeted support during off-peak periods can both improve rider experience and reduce operational losses.

Thank You.